
Short-Term Rental Prohibitions and Census-Tract Rents

A staggered difference-in-differences design with cohort-aware estimation

Data Science Institute Housing Clinic City of Chicago DTI
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What we ask, and why it is hard

- **Question.** Do census-tract STR prohibitions adopted under Chicago's Shared Housing Ordinance change *rental prices* for long-term tenants?
- **Why hard:** treatment is *staggered* – different tracts adopt in different months – and effects plausibly vary by cohort and time-since-adoption.
- **Implication for design:** pooled two-way fixed effects (TWFE) implicitly compares newly-treated tracts to *already-treated* ones, generating negative weights and biased ATT under heterogeneity (Goodman-Bacon, 2021).
- **Our spine:** the Callaway–Sant'Anna (CS) group-time ATT estimator with *never-treated* controls; pretrend matched sample; ACS + tract linear trend residualization for robustness.

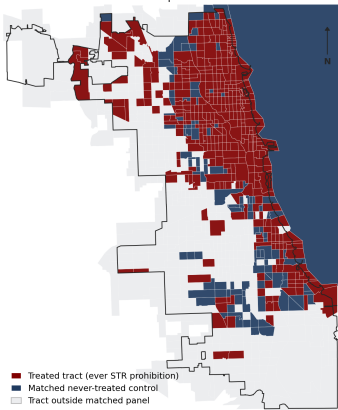
- **Outcome:** ZORI smoothed monthly rent index, ZIP-level, crosswalked to census tract using HUD area shares – a tract-month panel of rents in dollars.
- **Treatment:** first month a tract has *any* parcel covered by an STR prohibition (Chicago Data Portal: prohibited buildings).
- **Sample:** restricted to tracts whose pre-treatment rent slope can be matched to never-treated comparators (k-NN on slope, $k = 3, \geq 6$ pre-periods).
- **Result (after Steps 2–5):** matched panel = 373 treated tracts + 114 never-treated; CS keeps 12 prohibition-*cohort months* each with ≥ 5 tracts (covers 320 of the 373 treated tracts — very small rollout months are pooled out of aggregated curves).

Estimands operate on the rent index in \$/month; magnitudes are differences in rent levels, not in growth rates.

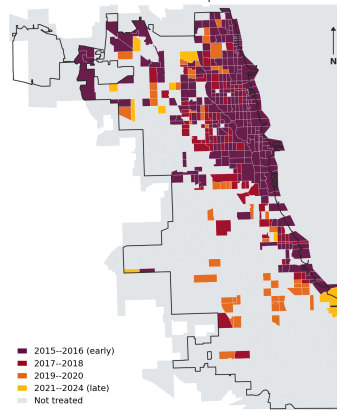
Geography of treatment (matched DiD panel)

Chicago STR-prohibition study: matched DiD sample (census tracts)

Who is in the matched DiD panel



When did treated tracts first see prohibition?



Sample: 487 matched tracts — 373 ever-treated, 114 never-treated comparators. Tracts outside the matched panel are shown in light grey.

Step 1 — Build one tract-month panel (before matching)

Audience hook: “What data did we actually start from?”

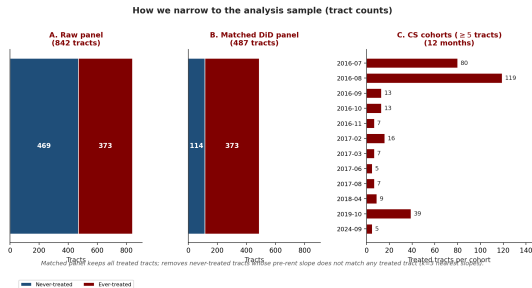
1. **Outcome:** smoothed rents (ZORI) measured at ZIP, allocated to census tracts via HUD ZIP–tract weights.
2. **Treatment timing:** first calendar month each tract intersects prohibited-building STR registrations (Chicago Data Portal).
3. **Panel bookkeeping:** one row per tract per month — e.g. 842 Chicago tracts in the raw file; aligned to 128 months (2015-01 through 2025-08).

Still no causal estimator — uniform panel structure is prerequisite for every subsequent plot.

Step 2 — Why shrink controls? (*slope matching*)

- Compared neighbourhoods differ in baseline prices; we nonetheless need untreated tracts whose *pre-prohibition monthly rent trajectory* resembles treated tracts' trajectories as closely as feasible.
- **Procedure:** for each treated tract, retain up to three never-treated tracts with closest pre-period rent slopes ($k=3$ NN in slope space); require ≥ 6 pre months.
- **Treated preservation:** *all* prohibition tracts survive this filter; removals apply only to never-treated tracts that never serve as usable counterfactual slopes.

Step 3 — Tract funnel (counts)



Stage	Description	Total tracts	Treated	Never-treated
A	Raw Chicago tract×month panel (ZIP rent crosswalk; prohibited-building timing). Before slope matching.	842	373	469
B	Matched DiD panel: retain never-treated tract if pre-treatment rent slope is among the nearest $k=3$ to at least one treated tract (≥ 6 pre-months).	487	373	114
C	CS-visible adoption months (≥ 5 treated). 12 cohort months; 320 treated tracts in tables/figures. (Excludes 53 treated tract(s) whose adoption-month cohort has < 5 units in pooled CS).			

Step 4 — Descriptive snapshots by prohibition cohort

Cohort	<i>n</i>	Share	Rent	Inc.	Bach.%	Rented%
2016-07	80	0.21	1,627	—	—	—
2016-08	119	0.32	1,340	—	—	—
2016-09	13	0.03	1,434	—	—	—
2016-10	13	0.03	1,447	—	—	—
2016-11	7	0.02	1,378	—	—	—
2017-02	16	0.04	1,282	—	—	—
2017-03	7	0.02	1,481	—	—	—
2017-06	5	0.01	1,312	—	—	—
2017-08	7	0.02	1,356	—	—	—
2018-04	9	0.02	1,445	—	—	—
2019-10	39	0.10	1,034	—	—	—
2024-09	5	0.01	1,305	—	—	—

- “Cohort” = calendar month each tract appears in the prohibition layer for the first time (group-time CS inputs).
- Early prohibition waves correlate with materially higher ACS incomes / baseline rents; later waves are geographically thinner.

Step 5 — Slopes aligned, levels still tilted (*residualization*)

- Matching enforces similarity in *rates of change*; it does **not** eliminate level wedges (cheap vs. expensive neighbourhoods remain cheap vs. expensive).
- Residualized specification:** fit on *pre-treatment months only*,

$$\hat{Y}_{it} = \hat{\alpha} + \hat{\gamma}' X_i^{ACS} + (\text{tract-specific linear } t),$$

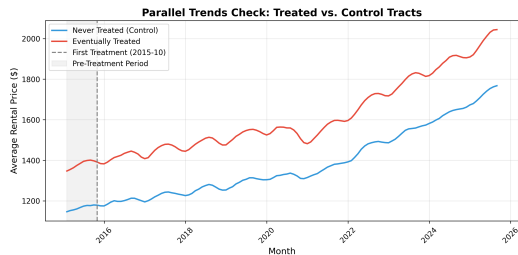
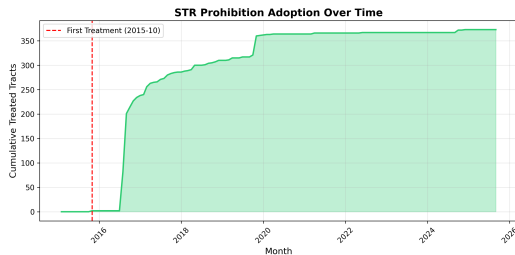
- Re-run pooled CS on $\tilde{Y}$ (drc) — magnitudes relocate because subtracting ACS + slopes changes the indexing of dollars.

Pre-window mean rents (months < first prohibition)			
Group	Mean	Std	Tract
Ever treated	\$1,379.67	313.86	37
Never treated	\$1,165.59	252.18	11

$t\text{-gap} \approx \$214, p < 10^{-11}$ (did_descriptive_pre_balance)

$\tilde{Y}_{it} = Y_{it} - \hat{Y}_{it} + Y$
Interpretation for policy slides: even “matched” contrasts mix different rent levels — hence transparent raw vs. residualized CS ladders.

Adoption dynamics and parallel rents

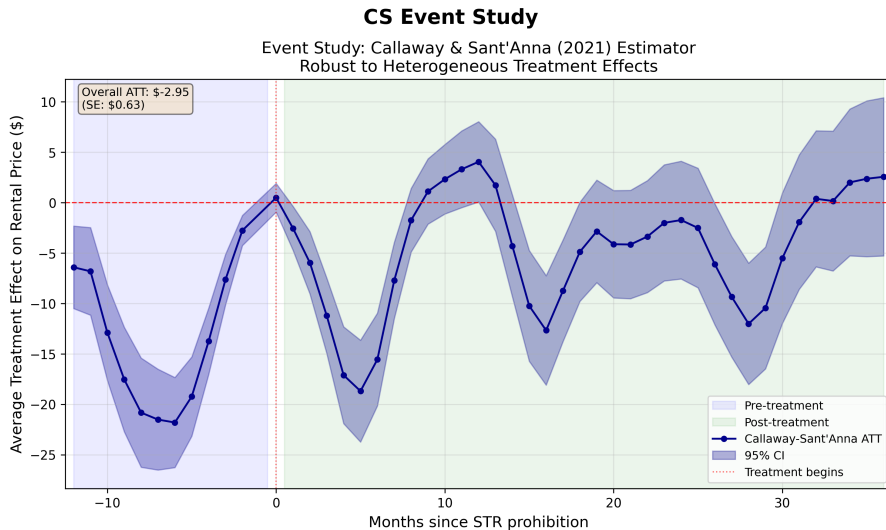


Rent paths visually parallel pre-prohibition; the level gap motivates residualization.

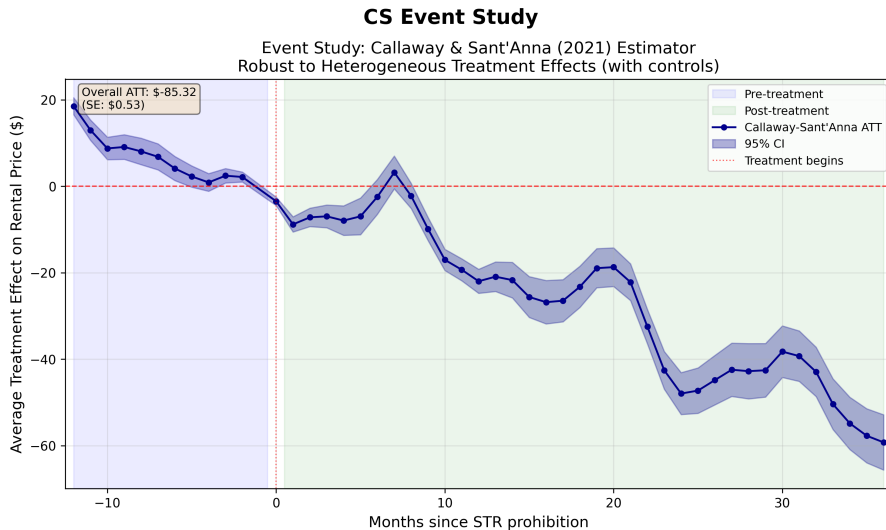
Why Callaway–Sant’Anna (and not pooled TWFE)

- For each cohort g and calendar period t , the CS estimator forms $\widehat{\text{ATT}}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G_g = 1] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$, comparing only to *never-treated* (or not-yet-treated) units.
- Aggregation to event-study time e : $\widehat{\text{ATT}}(e) = \sum_g \omega_{g,e} \widehat{\text{ATT}}(g, g + e)$ with positive, transparent weights.
- **What this buys us:** no contamination from already-treated controls, no negative weights, and an interpretable per-cohort dynamic effect.
- **Doubly robust analogue (drc):** residualize Y on pre-period ACS covariates and tract-specific linear trends, then re-run CS. Aligns with Sant’Anna and Zhao (2020).

Baseline CS event study (no covariate adjustment)



CS event study with ACS + tract-linear-trend controls



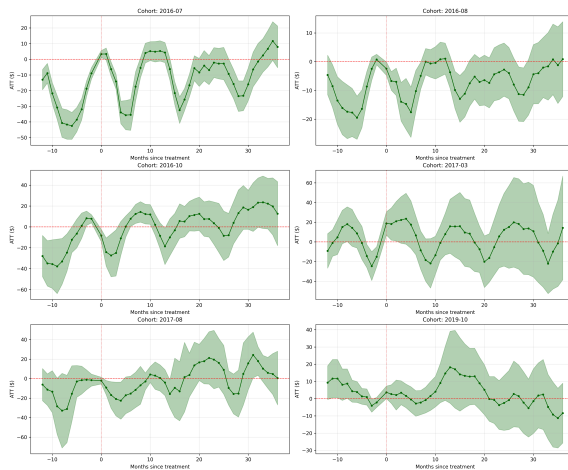
Pooled ATTs: baseline vs. residualized CS

Estimator	ATT (\$/mo.)	SE	95% CI
Callaway–Sant’Anna (matched sample)	−2.95	0.63	[−4.19, −1.71]
CS w/ ACS + tract linear trends (drc)	−85.32	0.53	[−86.37, −84.27]

- Both ATTs are statistically distinguishable from zero; the residualized estimate is interpretable on the residualized scale.
- The gap reflects the level wedge documented earlier: residualization sweeps out tract-specific drift that the unconditional CS leaves in.

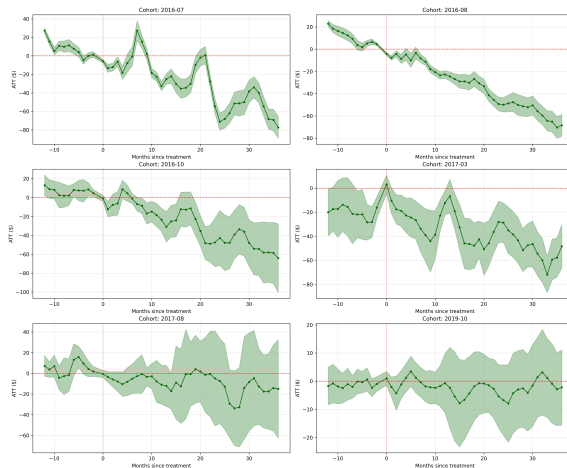
Cohort heterogeneity (baseline CS)

Cohort Dynamics

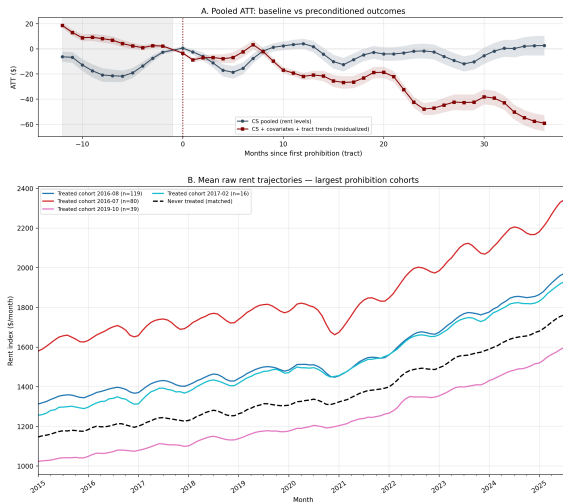


Cohort heterogeneity (CS with controls)

Cohort Dynamics

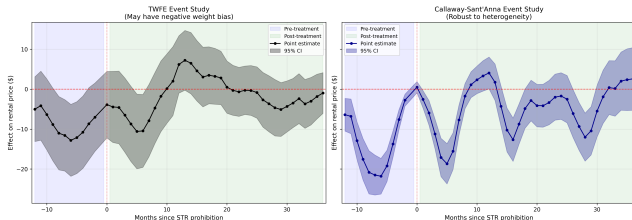


Why residualization moves the headline



TWFE vs. CS: is the bias guarantee binding here?

TWFE vs CS Comparison



Why?

- TWFE uses *already-treated* units as controls for newly-treated ones (“forbidden comparisons”) $\Rightarrow$ negative weights, biased ATT under heterogeneity.
- CS only compares to *never-treated* controls – no negative weights by construction.

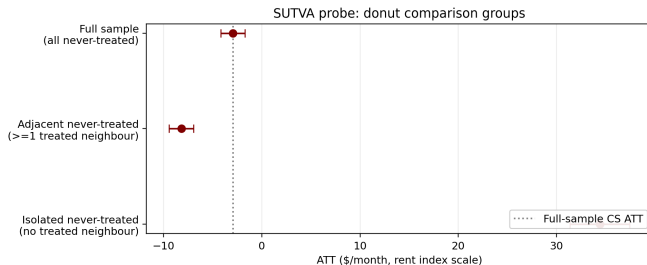
Takeaway

- Trajectories look parallel; mean TWFE–CS gap is \$2.83.
- TWFE bias is *not* the binding constraint in this Chicago sample, but we still report CS because its guarantee is unconditional.

Are controls contaminated by neighbour treatment?

- SUTVA failure $\Rightarrow$ never-treated tracts that border prohibition tracts may absorb spillover demand and bias the comparison.
- **Two probes:**
 1. *Donut*: restrict the never-treated pool to tracts with *no* ever-treated neighbour, then to those with at least one.
 2. *Dose*: sort never-treated tracts into quartiles of mid-sample share-of-neighbours-treated; re-estimate pooled CS in each.

Donut comparison: testing for spillover into controls



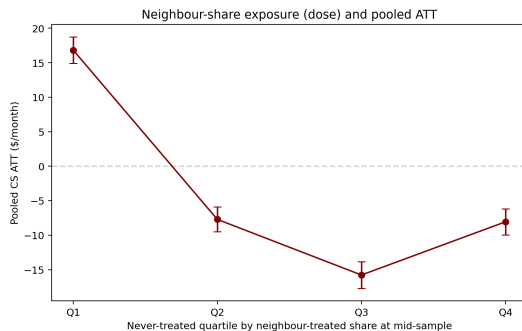
Idea

- Neighbours = tracts that share a border (geom.touches, queen contiguity).
- Split never-treated controls into *isolated* (no treated neighbour) vs *adjacent* (≥ 1 treated neighbour).
- Spillover hypothesis: adjacent controls absorb demand $\Rightarrow$ ATT pulled toward zero.

What we are seeing

- Full $-\$2.95$; Adjacent $-\$8.18$ ($n=100$); Isolated $+\$34.40$ ($n=14$).
- Clean SUTVA failure $\Rightarrow$ isolated should be *more* negative; we see the opposite.
- 14 isolated tracts are peripheral neighbourhoods $\Rightarrow$ selection, not spillover, drives the gap.

Neighbour-treatment dose



Construction. Sort never-treated controls by mid-sample share of treated neighbours; re-run pooled CS in each quartile.

- Q1 (share 0.13, $n=31$): +\$16.83.
- Q2 (share 0.44, $n=32$): -\$7.71.
- Q3 (share 0.66, $n=26$): -\$15.77.
- Q4 (share 0.92, $n=25$): -\$8.05.

Why we are seeing this

- Q1 mirrors the donut “isolated” subset: low-exposure controls sit in spatially distinct neighbourhoods.
- Q2–Q4 settle into a stable negative band $\Rightarrow$ no monotone dose gradient; pattern reads as neighbourhood selection rather than dose-dependent spillover.

Pretrend sensitivity (heuristic)

Diagnostic	Value
Max TWFE coef. in pre-periods	12.816
Max second difference of pre-coefs	3.086
Any strictly positive pre-coef.?	No

- Reports the maximum pre-period TWFE coefficient and the coarsest discrete second-difference (a smoothness budget) over pre-coefs.
- *Spirit only*: a calibrated bound under the Rambachan–Roth (2023) framework would replace these heuristics in a fully “honest” sensitivity exercise.

What we conclude

- **Tract-level STR prohibitions** are associated with a small *negative* change in tract rents on the unadjusted CS scale and a larger negative effect once tract-specific drift and ACS composition are netted out.
- **Design implication:** reporting both the “raw” CS and the residualized CS is honest: residualization changes the meaning of the ATT (it is now on the residual scale), not just its magnitude.
- **Robustness:** cohort heterogeneity is real; spatial probes do not flag obvious SUTVA failure; pretrend coefficients are small in magnitude.
- **Limits:** tract rents are interpolated from ZIP-level ZORI; “treatment” is binary even though prohibition density varies; no calibrated R&R bound is reported here.

- Callaway, B. and Sant'Anna, P. (2021). *Difference-in-differences with multiple time periods*. Journal of Econometrics 225(2), 200–230.
- Goodman-Bacon, A. (2021). *Difference-in-differences with variation in treatment timing*. JoE 225(2), 254–277.
- Sant'Anna, P. and Zhao, J. (2020). *Doubly robust difference-in-differences estimators*.
- Rambachan, A. and Roth, J. (2023). *A more credible approach to parallel trends*. RESTUD 90(5), 2551–2598.
- Pipeline source: `src/housing/scripts/did_pipeline_callaway_santanna.py`; documentation in `docs/CS_WITH_CONTROLS.md`.